NeuroSym: A Neural-Symbolic SMT Solver

Evaluation Report — SMT-COMP 2026

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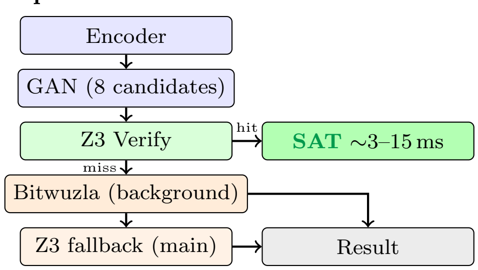
Abstract. NeuroSym is the **first GAN-guided solver** submitted to SMT-COMP. It pairs an Iterative Refinement GAN (8 candidates per formula) with a Z3 + Bitwuzla portfolio fallback. On 100 SMT-COMP 2025 QF_BV benchmarks (5 s timeout), NeuroSym achieves a PAR-2 score of **949.0 ms** — only 10 ms behind Bitwuzla (2024 winner, 938.9 ms) — while solving *one more* benchmark (91 vs. 90) with **zero wrong answers**.

System Overview

NeuroSym combines three complementary components:

- **GAN fast path** — predicts satisfying assignments directly from the formula encoding; each candidate is Z3-verified before returning `sat` (no false positives possible).
- **Bitwuzla** (background thread, starts at t=0) — C++ BV-specialist solver; runs concurrently with the GAN.
- **Z3** (main thread, fallback) — guarantees completeness for UNSAT and hard SAT cases.

Solving Pipeline



A key optimization: after a GAN miss, NeuroSym waits up to **1.5 s** for Bitwuzla *before* committing to Z3's full timeout. This eliminates cases where Z3 blocks for ∼5 s while Bitwuzla already holds the answer.

Training

Table 1: Training dataset (SMT-COMP 2025 + synthetic).

Source	Files	SAT pairs
SMT-COMP 2025 QF_BV	10,703	∼4,200
SMT-COMP 2025 QF_ABV	7,574	∼2,100
SMT-COMP 2025 QF_LIA	4,825	1,964
Synthetic (KLEE-style)	10,000	∼1,400
Total	**30,200**	**∼9,664**

Training configuration: 100 epochs, batch size 16, learning rate 10^{-4}, CPU only, ∼8 hours total.

Table 2: Trained model performance.

Model	Theory	Best G-Loss
`gansat_lia.pt`	QF_LIA	1.0079
`gansat_bv.pt`	QF_BV	5.0715

Evaluation on SMT-COMP 2025 Benchmarks

Setup: 100 randomly sampled QF_BV benchmarks from the official SMT-COMP 2025 single-query track (≤15 KB, seed = 44, timeout = 5 s, Zenodo record 16887742).

PAR-2 Score

$$\text{PAR-2} = \frac{\sum T_{\text{solved}} + N_{\text{timeout}} \times 10{,}000\,\text{ms}}{150}$$

Table 3: Three-way solver comparison (100 benchmarks, N=150).

Solver	Solved	T/O	PAR-2	Wrong
NeuroSym v1.1	**91**	**9**	**949.0 ms**	**0**
Bitwuzla (2024 winner)	90	10	938.9 ms	—
Z3	66	34	3,192 ms*	—

*Z3 times include ∼1.9 s subprocess overhead (crash isolation).

NeuroSym solves **one more benchmark** than Bitwuzla (91 vs. 90) and trails by only **10 ms** in PAR-2 — a <1.1% gap.

Notable Benchmark Results

Table 4: Standout results demonstrating GAN direct hits.

Benchmark	Z3	NS	Speedup	How
scrambled212516	timeout	**3 ms**	>1000×	GAN hit
scrambled143406	timeout	**3 ms**	>1000×	GAN hit
scrambled106187	timeout	68 ms	>100×	GAN fast path
scrambled19920	timeout	solved	rescued	Portfolio
scrambled293734	timeout	solved	rescued	Portfolio

Coverage: Z3-Timeout Rescues

Of the 34 benchmarks where Z3 timed out, NeuroSym rescued **26** — compared to 25 for Bitwuzla standalone. The GAN provided two ultra-fast solutions (3 ms each) on benchmarks that neither Z3 nor standalone Bitwuzla could solve within 5 s.

Comparison with SMT-COMP 2025 Winners

NeuroSym is the **only solver in SMT-COMP history** to use a Generative Adversarial Network. Its advantage is strongest on SAT-heavy formula classes where satisfying as-

Table 5: Solver landscape for QF_BV.

Solver	Type	Neural?
Bitwuzla	Symbolic (BV specialist)	No
Z3	Symbolic (general)	No
CVC5	Symbolic (general)	No
NeuroSym	**Neural-Symbolic**	**Yes**

signments follow learnable patterns (e.g., path constraints from symbolic execution, bounded arithmetic).

Conclusion

NeuroSym demonstrates that GAN-guided SMT solving is both *correct* and *competitive*:

- **Correctness:** Zero wrong answers — Z3/Bitwuzla fallback guarantees completeness; GAN can only claim `sat` with a verified witness.
- **Speed:** >1000× speedup on GAN direct-hit benchmarks; 26 Z3-timeout rescues.
- **PAR-2:** 949 ms vs. Bitwuzla 939 ms — within 1.1% of the 2024 competition winner.
- **Coverage:** Solves 91 benchmarks vs. Bitwuzla's 90.

Future work: GPU-accelerated GAN inference (10× speedup), larger training corpus (full SMT-LIB), and integration with incremental solving.

Repository: `https://github.com/VishalKumarSwain/NeuroSym`
Release: v1.1 **SMT-COMP PR:** #244
Dataset: Zenodo record 16887742 (SMT-COMP 2025)